



CSA1017 SOFTWARE ENGINEERING

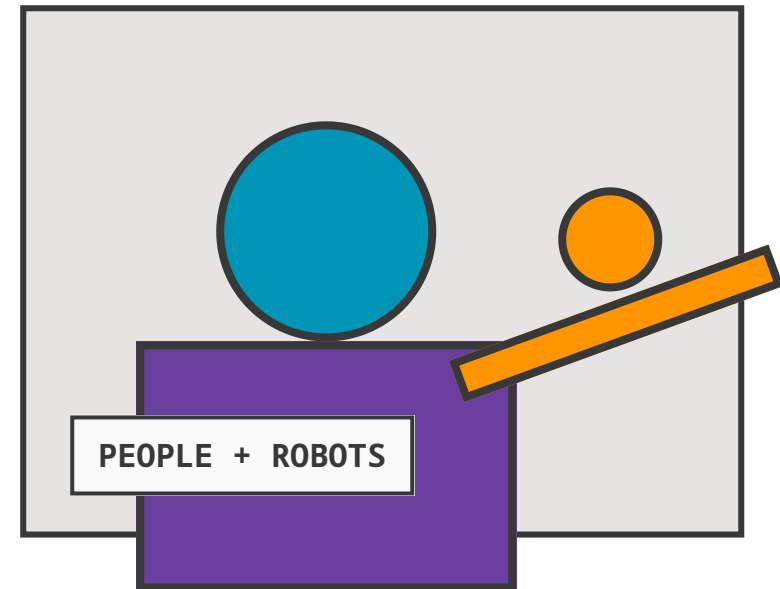
Human – Robot Interaction & Collaborative Robots

From isolated automation to safe, intelligent
teamwork.

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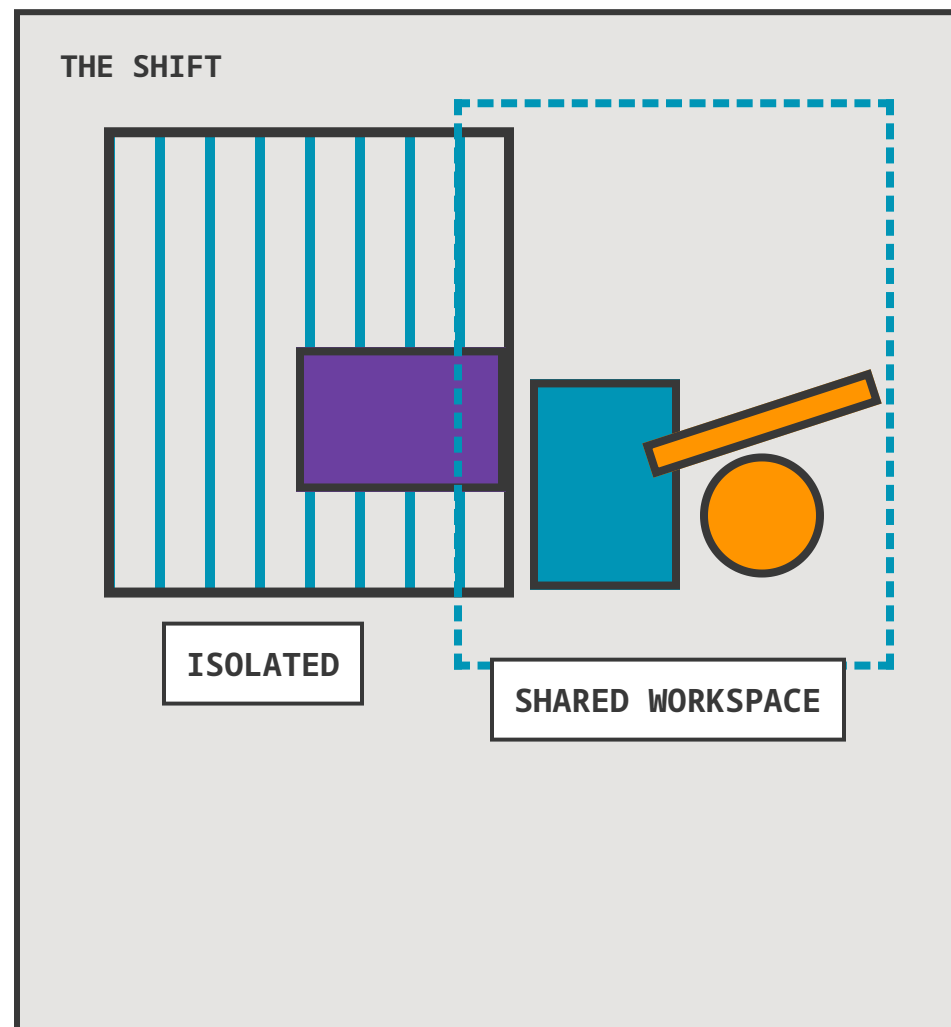
A systems view of sensing, software, safety, and human trust.



01 / INTRODUCTION

Robots move from cages to teams

- HRI designs how people perceive, communicate with, control, and collaborate with robots.
- **Cobots** are robot systems engineered for human–robot workspaces and workflows.
- The design goal expands from repeatability to **adaptability, safety, usability, and trust.**
- Software engineering connects sensing, interfaces, behavior, and safety into one cyber-physical system.





02 / NEED AND MOTIVATION

Why collaboration is the next automation layer

HUMAN STRENGTHS

- **Judgment:** Complex decision making and exception handling.
- **Dexterity:** Handling variable, soft, or delicate objects.
- **Context:** Awareness of the environment and changing goals.



ROBOT STRENGTHS

- **Precision:** Sub-millimeter accuracy in repetitive paths.
- **Endurance:** 24/7 operation without fatigue or injury risk.
- **Repeatability:** Consistent quality across thousands of cycles.

DESIGN GOAL: Allocate subtasks to the agent best suited to them.



03 / Working Principle

A cobot is a safety-controlled feedback loop

Every action is grounded in sensing, decision logic, controlled motion, and human-readable feedback.

01 PERCEIVE

Cameras, force/torque sensors, encoders, proximity sensing, and safety scanners observe people, objects, and robot state.

02 DECIDE

Task logic, motion planning, collision checks, and risk rules select the next safe action.

03 ACT

Controllers and actuators execute motion, grip, force, and speed commands within limits.

04 COMMUNICATE

UI, voice, gesture, haptics, and status signals keep the human informed and able to intervene.

SAFETY IS LAYERED

Physical

rounded links
compliant tooling

Control

speed & force
limits

Monitoring

separation
scanners

Operations

stop, assess,
recover

Engineering principle: safe behavior is a system property, not a single sensor feature.



04 / Key Technologies

The software stack turns motion into teamwork

APPLICATION LAYER

Task models, scheduling, digital twins, logging, analytics, and human-override policies.

INTERACTION LAYER

Teach pendants, hand-guiding, mixed reality, voice/gesture interfaces, and workload-aware feedback.

ROBOT MIDDLEWARE

Drivers, state estimation, kinematics, motion planning, and real-time control (e.g., ROS 2).

DEVICE LAYER

Motors, drives, encoders, end effectors, and safety-rated I/O hardware.

ENGINEERING PRIORITIES

- Deterministic Behavior
- Fault Containment
- System Traceability
- Simulation & Digital Twins
- Human-Centered Usability

Software engineering challenge:
Ensuring safety-critical performance
across non-deterministic human
environments.



05 / REAL-WORLD USE CASES

From factories to hospitals and homes

DOMAIN	EXAMPLE COLLABORATION	SOFTWARE FOCUS
Manufacturing	Assembly, machine tending, inspection, and shared screwdriving.	Skill libraries, vision systems, dynamic scheduling, safety zones.
Logistics	Order picking, kitting, palletizing, and shared material handling.	Perception, fleet coordination, exception recovery, SLAM.
Healthcare	Rehabilitation, lifting assistance, and medication handling.	Personalization, trust models, privacy, safe physical interaction.
Education	Robot programming, tutoring, and laboratory assistance.	Explainability, multimodal interaction, evaluation metrics.



06 / BENEFITS & LIMITS

Benefits are real—but never automatic

ADVANTAGES

- **Flexible Deployment:** Easily adapted for variable tasks and mixed human–robot workflows.
- **Better Ergonomics:** Offloads repetitive, heavy, or precision-intensive work from humans.
- **Consistent Quality:** Reliable execution and automated data capture for every cycle.
- **Accessibility:** No-code and natural-language interfaces lower the barrier for non-specialists.

LIMITATIONS

- **Safety Dependency:** A cobot is not inherently safe; safety depends on the *entire application*.
- **Reliability Gaps:** Perception errors and ambiguous human intent can reduce task success.
- **Lifecycle Costs:** Validation, training, maintenance, and cybersecurity add significant overhead.
- **Complexity:** Integrating dynamic human behavior into deterministic software is non-trivial.

CRITICAL: Safety is a system property verified through risk assessment, not a hardware feature.



07 / RECENT DEVELOPMENTS

The frontier is physical AI with measurable trust

INTELLIGENT SENSING

- **AI-Enhanced Perception:** Better recognition of objects, gestures, and human intent.
- **Uncertainty Awareness:** Robots that quantify and communicate their confidence levels.

SHARED AUTONOMY

- **Goal-Based Control:** Humans set high-level goals; robots handle low-level execution.
- **Mixed Reality (MR):** Spatial interfaces for intuitive teaching and debugging.

SKILL ACQUISITION

- **Learning from Demo:** Acquiring skills from human examples and simulation data.
- **Foundation Models:** Large-scale pre-training for cross-domain robotic tasks.

BENCHMARKING THE FUTURE

Shift from "impressive demos" to rigorous metrics for safety, workload, and acceptance.



08 / CHALLENGES AND FUTURE SCOPE

The hardest problems are social, technical, and systemic

SAFETY & VALIDATION

- Validate every task, tool, speed, and force against ISO/TS 15066.
- Dynamic risk assessment for changing human access paths.

INTENT & TRUST

- Communicate uncertainty to avoid overconfident, opaque behavior.
- Building explainable AI for robot-to-human handoffs.

HUMAN FACTORS

- Account for cognitive workload, fatigue, and accessibility.
- Adapting to changing user expertise and cultural norms.

DATA & SECURITY

- Protect sensor data, models, and robot controller pipelines.
- Ensuring cybersecurity in connected, AI-enabled workcells.

FUTURE SCOPE: Adaptive safety, certified learning systems, and swarm coordination.



09 / EXAMPLE APPLICATION

Case study: human-guided assembly cell

Scenario: Operator positions a variable part; cobot retrieves fasteners, presents the tool, and performs fastening.

- **01 Vision:** Confirms part identity and workspace state.
- **02 Teaching:** Operator uses hand-guiding to set the pose.
- **03 Planning:** Planner verifies separation and force limits.
- **04 Execution:** Cobot executes; torque sensing monitors load.
- **05 Recovery:** Robot detects fault, pauses, and requests help.

SYSTEM_STATUS.LOG

v1.0.4

```
> INITIALIZING PERCEPTION... OK
> TRACKING HUMAN_HAND... ACTIVE
> TORQUE_LIMIT: 15.0 Nm
> EXECUTING FASTEN_04...
```

```
> ERROR: ABNORMAL_TORQUE_DETECTED
> CAUSE: FASTENER_MISALIGNMENT
> ACTION: MOTION_PAUSED
> REQUEST: HUMAN_INTERVENTION_REQUIRED
```

ENGINEERING LESSON:

Graceful failure and clear handoff are as important as nominal task speed.



11 / SUMMARY & SOURCES

Design for collaboration, not just automation

The strongest cobot systems make human expertise more capable while keeping robot behavior understandable, interruptible, and accountable. Safety is a system property verified through risk assessment, simulation, and operational monitoring.

DESIGN PRINCIPLES

- 01 Task Decomposition: Identify complementary human-robot strengths.
- 02 Interaction Contracts: Define command, interrupt, and recovery roles.
- 03 Measurable Outcomes: Track safety, trust, and learnability.

AUTHORITATIVE REFERENCES

- [1] ISO/TS 15066:2016 — Collaborative Robots Safety Requirements.
- [2] OSHA Robotics Standards — Worker Protection Guidance.
- [3] NIST Collaborative Robotics — Performance & Benchmarking.